ANNIHILATION IN A FUSION PRODUCT FROM ANNIHILATION IN THE FACTORS

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Editorial note. Preamble provenance: the article keeps its theorem environments in the SIGMA publisher class and declares three operators of its own in the preamble of the pinned tex. This card therefore uses two verbatim preamble spans: the operator declarations from the pinned source (lines 9-11, required because the counted proof writes the associated graded with the operator declared there) and the staged retained-preamble.tex theorem-environment block (lines 1-28). Nothing was added, renamed or redefined, and extra_packages stays empty; the wrapper class and base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The fusion product of cyclic graded modules, defined through the associated graded of the tensor product with respect to the filtration by the powers of a parameter. The author's section 4.1, in which the fusion product is set up before the counted statement. Fixes the product in which the annihilation is asserted.
- The identification of the operator acting on the tensor product of the factors with the product of the shifted operators acting on the generators. Invoked inside the counted proof at source line 568 as a pointer to the lemma labelled f1 (a navigation-pointer stub). Computes the action of the twisted operator on the class of the tensor of the generators.

Source reference. Original reference f1: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 568–568. Navigation pointer only: the counted text refers to the item f1 of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

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Lemma 1. *Given $1 \leq i \leq m$, let V_i be a finite-dimensional cyclic graded $\mathfrak{g}[t]$-module generated by v_i, and $s_i \in \mathbb{Z}_{\geq 0}$. Let $x \in \mathfrak{g}$. If $(x \otimes t^{s_i})v_i = 0$, $\forall\, 1 \leq i \leq m$ then $(x \otimes t^{s_1+\cdots+s_m})v_1 * \cdots * v_m = 0$.*

Proof. Let $z_1, \ldots, z_m$ be distinct complex numbers and let $\mathbf{V}$ as above. By using Lemma f1, we get the following equality in $\operatorname{gr}\mathbf{V}$,

$$\begin{aligned}&\big(x \otimes t^{s_1+\cdots+s_m}\big)\big((v_1 \otimes \cdots \otimes v_m) + F^{-1}\mathbf{V}\big)\\ &= \big(\big(x \otimes (t-z_1)^{s_1} \cdots (t-z_m)^{s_m}\big)v_1 \otimes \cdots \otimes v_m\big) + F^{s_1+\cdots+s_m-1}\mathbf{V}.\end{aligned}$$

Now the proof follows by the definition of the fusion product. □